Learning Objective 2.3: I can install GNU Radio and use GNU Radio Companion (GRC) to build, inspect, and run a basic flowgraph.

Learning Objective 2.4: I can capture a real RF signal using an SDR and interpret its time- and frequency-domain representations.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Reading a flowgraph.** A classmate's flowgraph has a Signal Source, a Throttle, and a QT GUI Frequency Sink. GRC draws the arrow from the Throttle to the sink in red and will not run.

    (a) The source and throttle ports are blue and the sink's input is orange. What is wrong, and what are two ways to fix it?

    (b) The classmate then replaces the Signal Source with a Soapy RTLSDR Source and keeps the Throttle. Explain why the Throttle should come out.

2. **Complex sampling and aliasing.** A complex signal source runs at $f_s = 32$ kHz.

   (a) What range of frequencies can the samples represent?

   (b) Where does the frequency sink show a 40 kHz tone, and a -20 kHz tone?

   40 kHz:

   -20 kHz:

   (c) An RTL-SDR samples at 2.4 MS/s centered on 100.1 MHz. What band does the display show? A strong station at 102.1 MHz leaks past the RTL-SDR's filters unattenuated. Where does it appear?

   band:

   alias:

   (d) Explain why complex samples show twice the bandwidth of real samples at the same rate.

3. **Recording and reading IQ.** A File Sink records complex samples.

   (a) Find the size of a 5 s capture at 2.4 MS/s.

   (b) You analyze the capture with a 4096-point FFT. Find the width of one bin.

   (c) A teammate sends only the file `capture.cfile`. Name the two numbers you need before the file is useful, and explain why.

4. **Gain and dynamic range.** Each bit of an analog-to-digital converter adds about 6 dB of dynamic range.

   (a) Estimate the dynamic range of the RTL-SDR's 8-bit converter and the PlutoSDR's 12-bit converter.

   RTL-SDR: ____

   PlutoSDR: ____

   (b) With the gain at maximum, copies of a strong station appear at frequencies where nothing transmits. Name the problem and the fix.

   (c) A student tunes the RTL-SDR exactly onto a weak narrowband signal and finds it hidden in a spike at the center of the display. Explain the spike and the fix.

Documentation: _____